

1: Input and Type Conversion

Write a program that takes two inputs from the user: one string and one number.

- Print both values and their types.
- Convert the number to a float and print its type.

2: Conditional Grading System

Write a program that takes a student's mark as input and prints the grade using if-elif-else.

- A+ for marks ≥ 90
- A for marks ≥ 80
- B+ for marks ≥ 70
- Else: Failed

3: Sum of Digits

Write a function `sum_of_digits(n)` that calculates and prints the sum of digits of a given number using a while loop.

4: Menu-Driven Calculator

Create a menu-driven calculator using a while loop.

Options:

1. Add
2. Subtract
3. Multiply
4. Divide
5. Exit

Use continue for invalid choices and break to exit.

5: List Comprehension Practice

Use list comprehension to generate:

- A list of squares from 1 to 10
- A list of even numbers from 1 to 20
- A list of characters in the string "Python"

6: Search in List

Write a program to search for a number in a list using a for loop.

- If found, print "Found!" and break.
- If not found, print "Not found" using the else clause of the loop.

7: Function with Variable Arguments

Write a function that accepts any number of positional and keyword arguments.

- Print the type and contents of both args and kwargs.

8: Global vs Local Variables

Write a program that demonstrates the difference between global and local variables.

- Use a global variable and modify it inside a function using the `global` keyword.

9: ASCII Conversion

Write a program that:

- Takes a character input and prints its ASCII value using `ord()`
- Takes an ASCII value and prints the corresponding character using `chr()`

10: Range and Loop Patterns

Use `for` and `while` loops to print:

- Numbers from 1 to 10
- Odd numbers from 1 to 20
- A pattern: 3, 7, 11, ..., up to a user-defined limit